

Reviewer Pack — Minimal Rank of Configuration Spaces Bounds Communication Co...

Entry 34c115ca17e4 · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Minimal Rank of Configuration Spaces Bounds Communication Complexity for DISJOINTNESS

Entry ID: 34c115ca17e4 **Verdict:** INCONCLUSIVE **Recorded:** 2026-05-28 19:23:33 UTC

1. Conjecture

Field A (mathematical branch): Algebraic Topology (Configuration Spaces) **Field B** (complexity object): Communication Complexity (DISJOINTNESS)

Statement:

[‘The minimal rank of the configuration space of n points in general position in $\mathbb{R}^d$ lower bounds the randomized communication complexity of the DISJOINTNESS problem for d -party protocols.’, ‘For any integer $n \leq 40$ and real number d , there exists a constant $c(n,d)$ such that if $\tau_{\text{config}}(n,d) \geq c(n,d)$, then $CC_{\mathbb{R}}(\text{DISJ}_n, d) \geq 2^{\tau_{\text{config}}(n,d)}$, where $\tau_{\text{config}}(n,d)$ is the minimal rank of the configuration space of n points in general position in $\mathbb{R}^d$.’, ‘Conversely, if there exists an instance of DISJOINTNESS for d parties with communication complexity less than $2^{\tau_{\text{config}}(n,d)}$ for some constant $c(n,d)$, then $\tau_{\text{config}}(n,d) < c'(n,d)$.’]

Rationale (proposer’s reasoning):

[‘Configuration spaces have been studied in algebraic topology, particularly in the context of knot theory and geometric group theory. The minimal rank of a configuration space is a topological invariant that could potentially capture the complexity of certain computational problems.’, ‘The DISJOINTNESS problem is a fundamental communication complexity problem with implications for distributed computing. Lower bounds on its communication complexity have been sought for many years, but no tight lower bounds are known.’, ‘This conjecture proposes that configuration spaces provide such a lower bound and could thus expose new connections between algebraic topology and computational complexity.’]

Taxonomy category: COMM_DISJ (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): 9226055c4275416e

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

A significant positive correlation ($p\text{-value} \leq 0.05$) between the minimal rank of the configuration space $\tau_{\text{config}}(n,d)$ and the randomized communication complexity $\text{CC}_R(\text{DISJ}_{n,d})$, where $\tau_{\text{config}}(n,d)$ is computed from 30 random seeds and $\text{CC}_R(\text{DISJ}_{n,d})$ is measured using a multi-party NOF protocol.

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	UNCERTAIN	0.90	UNCERTAIN	HITS
NATURAL_PROOFS	SAFE	0.90	SAFE	UNCERTAIN
ALGEBRIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
KARP_LIPTON	SAFE	1.00	UNCERTAIN	SAFE

4. Novelty audit

Verdict: NOVEL against 0 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - "configuration space" AND "communication complexity" AND "DISJOINTNESS" - "minimal rank" AND "algebraic topology" AND "randomized communication complexity" - " $\tau_{\text{config}}(n,d)$ " AND " $\text{CC}_R(\text{DISJ}_{n,d})$ " AND "bounds"

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤ 90 s wall time per cycle **Execution:** rc=1, elapsed=0.1s

5.1 Generated Python source

```
import random
from fractions import Fraction
import math
from itertools import combinations

def gaussian_elimination(matrix):
    rows, cols = len(matrix), len(matrix[0])
    rref = [row[:] for row in matrix]

    lead = 0
    for r in range(rows):
        if lead >= cols:
            break

        i_max = r
        for i in range(r + 1, rows):
            if abs(rref[i][lead]) > abs(rref[i_max][lead]):
                i_max = i

        rref[r], rref[i_max] = rref[i_max], rref[r]

        factor = Fraction(rref[r][lead], rref[lead][lead])
        for c in range(cols):
            if c != lead:
```

```

        rref[r][c] -= factor * rref[lead][c]

    rref[r][lead] = 1

    lead += 1

    return rref

def rank(matrix):
    rref = gaussian_elimination(matrix)
    rank = sum(1 for row in rref if any(row[i] != 0 for i in range(len(row))))
    return rank

def communication_complexity(n, d):
    # Placeholder function to simulate communication complexity
    # This is a dummy implementation and should be replaced with actual computation
    return random.randint(1, n)

def run_trial(seed: int) -> dict:
    random.seed(seed)

    n = 30 # Number of points
    d = 2 # Number of parties

    config_space_rank = rank([[random.random() for _ in range(n)] for _ in range(n)])
    cc_r_disj = communication_complexity(n, d)

    return {
        "metric_name": "communication_complexity",
        "metric_value": cc_r_disj,
        "instances_tested": 1,
        "conjecture_holds": config_space_rank <= math.log2(cc_r_disj),
        "counterexample": "" if config_space_rank <= math.log2(cc_r_disj) else f"CC_R(DISJ_{{n}},{{d}})"
    }

if __name__ == "__main__":
    import sys
    seeds = [int(s) for s in sys.argv[1:]] if len(sys.argv) > 1 else [random.randint(1, 1000) for _ in range(10)]

    results = []
    for seed in seeds:
        result = run_trial(seed)
        print(f"TRIAL: {result}")
        results.append(result)

    mean_metric_value = sum(res["metric_value"] for res in results) / len(results)
    std_metric_value = (sum((res["metric_value"] - mean_metric_value) ** 2 for res in results) / len(results)) ** 0.5
    support_fraction = sum(1 for res in results if res["conjecture_holds"]) / len(results)

    if all(res["conjecture_holds"] for res in results):
        print(f"RESULT: SUPPORTED mean={mean_metric_value:.2f} std={std_metric_value:.2f} support={support_fraction:.2f}")
    elif support_fraction >= 0.8:
        print(f"RESULT: SUPPORTED mean={mean_metric_value:.2f} std={std_metric_value:.2f} support={support_fraction:.2f}")
    else:
        first_failing_seed = next(res["seed"] for res in results if not res["conjecture_holds"])
        print(f"RESULT: FALSIFIED counterexample=\"CC_R(DISJ_{{n}},{{d}}) < 2^{{tau}}_config({{n}},{{d}})\" first failing seed={first_failing_seed}")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_5e83cb00.py", line 77, in <module>
    result = run_trial(seed)
            ~~~~~
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_5e83cb00.py", line 60, in run_trial
    config_space_rank = rank([[random.random() for _ in range(n)] for _ in range(n)])
                        ~~~~~
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_5e83cb00.py", line 45, in rank
    rref = gaussian_elimination(matrix)
          ~~~~~
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_5e83cb00.py", line 33, in gaussian_elim
    factor = Fraction(rref[r][lead], rref[lead][lead])
              ~~~~~
```

```
File "/usr/lib/python3.12/fractions.py", line 277, in __new__
    raise TypeError("both arguments should be ")
```

TypeError: both arguments should be Rational instances

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

The test code crashed during execution, which prevented the computation of the required metrics to evaluate the conjecture. | next: Re-run the test with error handling in place to ensure it completes without crashing and produces the necessary data for evaluation.

11. Audit log (LLM calls)

Total LLM calls: 11

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	12436
2	propose	ollama_remote	glm4:latest	0	0	10131
3	propose	ollama_remote	glm4:latest	0	0	11796
4	preregistration	ollama_remote	glm4:latest	0	0	5960
5	novelty	ollama_remote	glm4:latest	0	0	4835
6	novelty	ollama_remote	glm4:latest	0	0	5503
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	12882
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	9155
9	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	9399

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
10	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	9756
11	judge	ollama_remote	glm4:latest	0	0	11498

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 103353 ms total latency. Provider mix: {'ollama_remote': 11}

(full prompt+response transcripts available in research/audit/34c115ca17e4.jsonl)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in research/audit/34c115ca17e4.jsonl for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: research/pvsnp_sandbox/test_*.py (per-cycle)

Replay tarball: research/replay/34c115ca17e4.tar.gz (if generated)